

Roll No.

TCS-342

B. TECH. (CSE) (AI & DS) (THIRD SEMESTER)

MID SEMESTER EXAMINATION, Oct., 2022

INTRODUCTION TO STATISTICAL DATA SCIENCE

Time : 1½ Hours

Maximum Marks : 50

Note : (i) Answer all the questions by choosing any *one* of the sub-questions.

(ii) Each sub-question carries 10 marks.

1. (a) Discuss data science life cycle. (CO5)

OR

(b) Discuss Poisson and Normal distribution with its mean, variance and standard deviation. (CO5)

2. (a) The probability that a laptop manufactured by a company will be defective is $1/10$. If 15 such laptops are manufactured, apply binomial distribution to find the probability that : (CO2)

(i) Exactly two will be defective

(ii) At most three will be defective

(iii) At least three will be defective

P. T. O.

OR

- (b) It is known that past experience in a certain plant there are on the average of 5 industrial accidents per month. Apply Poisson distribution to find the probability that in a given year there will be : (CO2)

- (i) Exactly 2 accidents
- (ii) At most 3 accidents
- (iii) More than 4 accidents

3. (a) Discuss linear and multi-linear regression. (CO6)

OR

- (b) The saving bank account of a customer showed an average balance of \$150 and a standard deviation of \$50. Assuming that account balances are normally distributed, find what percentage of account is : (CO6)

- (i) over \$200
- (ii) between \$120 and \$170
- (iii) less than \$75

Given that :

$$P_{(0 < Z < 1)} = 0.3413$$

$$P_{(Z < 0.4)} = 0.6554$$

$$P_{(Z < -0.6)} = 0.2743$$

$$P_{(0 < Z < 1.5)} = 0.4332$$

4. (a) Marks obtained by 5 students in algebra and trigonometry are as given below : (CO1)

Algebra	Trigonometry
16	11
15	18
12	10
10	20
8	17

Calculate the Karl Pearson's correlation coefficient.

OR

- (b) Calculate rank correlation from the following data related to a group of 10 students and percentage of marks secured : (CO1)

Roll No.	% of marks in Science	% of marks in Mathematics
1	45	70
2	66	81
3	55	75
4	45	75
5	80	70
6	75	85
7	50	65
8	55	80
9	60	45
10	45	60

P. T. O.

5. (a) Elaborate the various types of correlation. (CO4)

OR

- (b) The heights (in feet) of the different persons are follows : (CO4)

1.2, 2.3, 4.9, 5.1, 5.2, 5.8, 5.9, 6.2, 6.5, 7.1, 24.5, 33.2, 40.4

Find heights outlier by applying IQR method.

Roll No.	% of marks in Science	% of marks in Mathematics
1	45	70
2	66	81
3	22	75
4	45	75
5	80	70
6	75	82
7	20	65
8	22	80
9	60	45
10	45	60